# Discrete Drainage Dynamics II: Cascade-Throttled Drainage and a Schwarzschild-Like Time-Dilation Factor

Stanislas Dewavrin[1]

[1]Independent researcher , dewavrin.iphone@gmail.com

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**Abstract**

**Discrete Drainage Dynamics (DDD) is an analogue-substrate programme.** It does not contest General Relativity, the Newtonian limit, or the Standard Model: those are empirical anchors that any analogue must reproduce in the appropriate continuum and regime limits. DDD asks the more modest, exploratory question of whether some of the *forms* that appear fundamental in current physics can be made mechanically legible inside a deliberately simpler discrete toy substrate. The present paper studies the static spherical gravitational sector of that analogue.

Paper I introduced DDD as a reserve-flux framework with a mobility function $\mu(R)$ that the framework left open, and studied the constant-mobility baseline $\mu \equiv 1$ that recovers the form of Newton's $1/r$ at leading order in the continuum limit of the analogue substrate. Here we close that gap. Within the class of local emission-capacity rules, we show that four operational axioms — locality, internal conservation, empty-reserve extinction, and additivity of independent reserve packets (L4) — uniquely select the source-side mobility $\mu(R) = R/R_0$ via the Cauchy functional equation and standard regularity. *The construction rests on two substantive hypotheses not derived from deeper principles: (H1) the additivity axiom (L4), and (H2) the bandwidth identification $d\tau/dt = R/R_0$ of effective proper time with the reserve-normalised bandwidth (introduced in Sec. 3). Under these two hypotheses and the locality axioms (L1)–(L3), the source-side mobility is forced and the Schwarzschild form follows; the microscopic origin of (H1) and (H2) is the principal open problem of the construction and is left to future work.* The directional flux on a link is therefore modulated by the bandwidth available at the source node, with no receiver-side throttle in the emission-capacity sector. Under this axiomatically selected mobility, the steady-state field around a point sink satisfies the algebraic identity $\nabla^2(R^2) = 0$ in the continuum limit, away from the sink. The unique spherically-symmetric solution with $R \to R_0$ at infinity is $R/R_0 = \sqrt{1 - \beta/r}$, with $\beta$ fixed by the source strength. This is the form of the Schwarzschild static clock-rate factor, $\sqrt{1 - 2GM/c^2 r}$, under the identification $\beta = 2GM/c^2$. Combined with the operational identification of effective proper time with local bandwidth ($d\tau/dt = R/R_0$), the model reproduces a Schwarzschild-like static clock-rate factor in the continuum spherical solution, conditional on the additive emission-capacity axioms and the bandwidth identification. We verify the result numerically. From the same structural input we extract the functional form of Newton's gravitational constant $G$ in terms of the local rule parameters $(\kappa, \alpha, R_0)$ and the lattice scale, together with a single Planck-anchored dimensionless constraint $\kappa/(\alpha R_0) = 4\pi$. *This Planck-anchoring condition is a consistency requirement under the Planck identification, not an independent prediction of $G$.* We do not predict the SI value of $G$ (the absolute lattice scale must be anchored externally), nor do we derive the speed of light $c$, the spatial part of the metric, or the Einstein field equations.

**What this paper makes mechanically legible.** GR predicts the Schwarzschild form $g_{tt} \propto -(1-\beta/r)$ with metric precision but supplies no mechanism for *why* clocks tick slow near a mass, and produces a curvature singularity at $r = 0$ where the continuum theory breaks down. *This paper* reads $g_{tt}$ as a substrate-level bandwidth ratio $R/R_0$ under source-side throttling: nodes near a sink literally have less bandwidth available, slowing their internal event rate. The would-be singularity is replaced by a *bandwidth horizon* where Paper I's discrete rate-limiter activates. **Leverage:** the form of $g_{tt}$ acquires a mechanism, and GR's curvature singularity is regularised by the discrete substrate — the marker that the analogue can do better than the continuum where the latter signals its own breakdown.

# 1 Main claims and scope

**Programme positioning and analogue framing.** Before stating the claims of this paper we recall the position of the entire DDD programme. DDD is an analogue-substrate programme: General Relativity, the Newtonian limit, and the Standard Model are not contested; they are the empirical anchors any analogue must reproduce in the appropriate continuum and regime limits. The aim of this paper is not to replace GR but to ask whether the *form* of the Schwarzschild static clock-rate factor can be made mechanically legible inside a deliberately simpler discrete substrate with finite reserves and local update rules. The results are internal to the analogue and conditional on the stated hypotheses (H1) and (H2); they do not claim to derive the underlying physical law. Any deviation between the analogue and the empirical theories is a falsifier of the analogue, not of the empirical theories.

1. **Bandwidth identification.** Within the rule of Paper I, the maximum directional flux a node $i$ can push out is $\alpha R_i$, set by the locally available reserve. We identify the dimensionless ratio $R_i/R_0$ with the local rate of effective proper time:

$$\frac{d\tau_i}{dt} = \frac{R_i}{R_0}.$$

   This identification is operational, not derived; it states that a node's local rate of activity per coordinate tick is proportional to its available reserve.

2. **Source-side bandwidth mobility, derived from four locality axioms.** Within the rule of Paper I, the local emission capacity $C(R)$ of a node is constrained by four operational axioms ((L1) locality, (L2) internal conservation, (L3) empty-reserve extinction, (L4) additivity of independent reserve packets). The Cauchy functional equation that (L4) imposes, together with the normalisations $C(0) = 0$ and $C(R_0) = 1$ from (L3) and rest-state calibration, has a unique continuous solution $C(R) = R/R_0$. The source-side bandwidth mobility is therefore *uniquely selected* within the class of local additive emission-capacity rules:

$$\mu(R) = \frac{R}{R_0}, \qquad F_{i\to j} = \alpha \frac{R_i}{R_0}(R_i - R_j)_+.$$

   The full derivation is in Sec. 4; alternative prescriptions discussed in Sec. 5.2 all violate at least one of (L1)–(L4).

3. **Schwarzschild form from the source-side bandwidth mobility.** In the continuum limit, the steady-state field of the rule with $\mu(R) = R/R_0$ around a point sink satisfies $\nabla^2(R^2) = 0$ away from the sink. The unique spherically symmetric solution with $R \to R_0$ at infinity is

$$\frac{R(r)}{R_0} = \sqrt{1 - \frac{\beta}{r}}, \qquad \beta = \frac{\kappa E_0}{2\pi\,\alpha\,R_0}.$$

Combined with the bandwidth identification, this gives the gravitational time-dilation factor

$$\frac{d\tau(r)}{dt} = \sqrt{1 - \frac{\beta}{r}},$$

which under the calibration $\beta = 2GM_{\text{phys}}/c^2$ matches the Schwarzschild static clock-rate factor as a formal function of $GM/c^2r$. The match holds in the continuum spherical solution, conditional on the additive emission-capacity axioms and the bandwidth identification; we do not claim a derivation of the full Schwarzschild metric or of weak-field GR more generally.

**Limitations and scope.**

We do not derive special relativity, and we do not address kinematic (motion-induced) clock-rate effects; those are deferred to Paper III. We do not derive the full Schwarzschild metric; we derive a $g_{00}$-like clock-rate factor for a static spherically symmetric source, in the continuum spherical solution. The spatial part of the metric, the non-spherical and non-static sectors, and the Einstein field equations are outside the scope of this paper. We do not predict the numerical value of Newton's gravitational constant $G$; in Sec. 9 we derive its functional form in terms of the local-rule parameters and a single Planck-scale dimensionless constraint, but the absolute lattice scale must be anchored by an external input. The speed of light $c$ enters as a calibration, exactly as in Paper I. We do not show that DDD's full local rule is unique; we show that the source-side mobility is uniquely selected within the additive local emission-capacity class. The mobility function $\mu(R)$ is *derived* from four locality axioms (Sec. 4): the unique continuous solution to the Cauchy functional equation imposed by additivity of independent reserve packets is $\mu(R) = R/R_0$. We do *not* predict the absolute value of $R_0$ in lattice units; that is an external calibration to the rest state of bound matter. Alternative prescriptions discussed in Sec. 5.2 (constant mobility, two-sided product) all violate at least one of the four axioms and are reported as null hypotheses; the Schwarzschild form is the unique consequence of the axiomatically forced mobility.

## Motivation and significance

The deepest conceptual feature distinguishing general relativity from Newtonian gravity is that *gravity itself gravitates*. In Newton's theory, the gravitational potential of two masses is the linear sum of two single-mass potentials: superposition is exact, the gravitational field is non-self-coupled, and the field carries no energy. In Einstein's theory, this picture breaks down: the gravitational field carries stress-energy, and stress-energy is the source of gravity, so the field couples to itself non-linearly. The Schwarzschild clock-rate factor $\sqrt{1 - 2GM/(rc^2)}$ is one of the simplest visible signatures of this non-linear structure: its expansion contains corrections beyond the Newtonian $1/r$ potential, reflecting the way the gravitational field's own energy contributes to the source of further field. (The metric component $g_{00}$ is linear in $1/r$ in Schwarzschild coordinates; the non-linear content sits in the clock-rate factor and in the relation between coordinates and proper observables.)

This is conceptually mysterious. *Where does the self-coupling come from physically?* If energy is what generates gravity, and the gravitational field has energy, then the field generates more of itself — but the back-reaction loop has no obvious mechanical substrate in standard general relativity. It is encoded in the non-linearity of Einstein's field equations as a mathematical identity, but the physical *ontology* of how gravity bootstraps itself remains opaque. Standard expositions either treat it as "geometry being non-linear" or invoke energy-momentum pseudo-tensors with their well-known coordinate ambiguities.

This paper proposes that the finite-resource substrate of Paper I provides exactly the missing ontology. Each node carries a *non-negative* reserve $R_i \geq 0$, and the local drainage rate at a node is regulated by its own reserve via a mobility function $\mu(R)$. When mass concentrates and

locally depletes the reserve, the reduced reserve directly slows the drainage at that node — and that slowing plays the role of the back-reaction that produces the Schwarzschild-like correction. The key statement:

> *Self-coupling of gravity in GR corresponds, in DDD, to self-throttling of drainage by the finite reserve. Where GR posits a non-linear field equation, DDD instantiates a non-linear self-back-reaction directly into the local conservative rule.*

Concretely, with the source-side bandwidth-throttled mobility $\mu(R) = R/R_0$, the steady-state radial profile satisfies a non-linear ODE whose solution is $R(r) = R_0\sqrt{1-\beta/r}$ with $\beta = 2GM/c^2$. The Schwarzschild square-root factor emerges not from a postulated geometry but from a finite-resource feedback: the same reserve that responds to mass-energy is the resource that gates its own response.

The mystery has therefore not disappeared, but been displaced into a sharper, simpler question: *why does the substrate carry a non-negative reserve with a regulated mobility?* That question is at least concrete — it is about the ontology of the substrate itself, not about the mathematical non-linearity of field equations. Papers III–VII address it from different angles (kinematics, Wilson loops, gauge sector, gravity-gauge coupling); Paper V provides the falsification windows.

**Which mystery does this paper address?** Per SERIES_FEEDBACK §0.bis, this paper targets:

- **Mystery #2 (gravitational time dilation and gravitational potentials).** GR encodes both in metric geometry and predicts the Schwarzschild form $g_{tt} \propto -(1-\beta/r)$. We ask whether the same form arises operationally from the analogue's source-side bandwidth-throttled rule. The answer offered here: under the hypotheses (H1) additivity (L4) and (H2) bandwidth identification, the analogue produces $R/R_0 = \sqrt{1-\beta/r}$ in the static spherical continuum limit, matching the form of the Schwarzschild factor. The microscopic origin of (H1) and (H2) remains open and is the principal *why*-question this paper does *not* resolve.

- **Mystery #4 (BH singularity).** The bandwidth profile reaches $R = 0$ at $r = \beta$, where the discrete rate-limiter of Paper I becomes active. The paper does not fully resolve the singularity but identifies the bandwidth horizon as the analogue's regularising structure; full treatment is deferred.

**Distinctive contribution and ontological reading.** GR predicts the Schwarzschild form $g_{tt} \propto -(1-\beta/r)$ with extraordinary precision. It does not, however, supply a *mechanism* for *why* time runs slow near a mass, nor does it resolve the curvature singularity at $r = 0$ that appears in its own predictions. These are the two distinctive points where the present paper's analogue contributes something the continuum theory does not say:

1. **Mechanical reading of gravitational time dilation.** In GR, $g_{tt}$ is a metric component; the *form* $\sqrt{1-\beta/r}$ is encoded in the geometry without further explanation. Here, the same form is the steady-state bandwidth ratio $R/R_0$ around a sink under the source-side throttled rule: *nodes near a mass simply have less bandwidth available, and this physically reduces the rate at which their internal events can occur.* The analogue provides an operational why for the form: throttle the substrate's local activity, and the Schwarzschild factor appears as a *consequence* of the discrete rule.

2. **Singularity regularisation: the marker that the continuum has a limit.** The Schwarzschild solution of GR has a genuine curvature singularity at $r = 0$ where the theory breaks down — this is widely regarded as one of GR's known *contradictions* or limits. In

the analogue, the bandwidth profile reaches $R = 0$ at $r = \beta$, where the *discrete rate-limiter* of Paper I activates and prevents $R$ from going negative. The would-be singularity is therefore replaced by a *bandwidth horizon*, a finite-substrate structure with no continuum-style divergence. **This is precisely the ontological marker that the discrete model can do better than the continuum model: where GR signals its own breakdown, the analogue stays well-defined by virtue of its finite-reserve structure.** Full treatment of the regularised horizon is deferred, but the mechanism is already in place.

**What is calibrated and what is genuinely explanatory.** The numerical value of $\beta = 2GM/c^2$ is calibrated against the Newtonian limit (see calibration ledger, §0.ter); $G$ in physical units is calibrated against CODATA. These values are *not* predicted by the analogue in this paper. What *is* predicted, structurally, is the *form* $\sqrt{1-\beta/r}$ as a substrate consequence, and the *mechanism* of singularity regularisation as a consequence of (L4) reserve non-negativity. The distinctive contribution is the displacement of two GR mysteries (why this form, what happens at $r = 0$) into substrate-level statements.

## 2 Physical picture in plain words

Picture each node as a tank holding a non-negative amount of stuff at level $R$, with a rest level $R_0$. The pipes between neighbours carry flow whose magnitude depends on the level difference. There is one limit on what any tank can deliver: you cannot push out more stuff than the tank actually holds. A full tank can sustain the rated flow $\alpha R_0$. A half-full tank can sustain at most $\alpha R_0/2$. An empty tank can deliver nothing. The rate at which a tank can engage with its environment is bounded by the level it currently holds, and not by anything else.

Now think about local time at the tank. The rule provides no hidden mechanism, no internal oscillator, no auxiliary phase. A tank's participation in the rule is entirely its flux activity. We take the rate of effective proper time at a tank to be the rate at which it can support that activity per coordinate tick. Full tank: rated rate. Half-full: half rate. Empty: no rate. Proper time *is* the throughput.

Drop a sink near a region of full tanks. The sink absorbs. Neighbouring tanks lose stuff to the sink. Their levels drop. Their throughput capacity drops. Their effective proper time slows down. The neighbours of the neighbours feel the same pull, and the cascade propagates outward through the lattice.

Now ask the question that gives the paper its central result. The cascade itself is throttled by bandwidth: a tank that has been drained can push less stuff than a full tank, even for the same level difference. This is the physically natural form of the rule — the source needs reserve to push, the receiver does not need anything to accept. So the flux on a link is not just proportional to $(R_i - R_j)$, it is proportional to $(R_i/R_0)(R_i - R_j)$: the source's bandwidth limits its own emission rate.

What does the cascade settle into? Far from the sink, levels are barely perturbed and the throttle is invisible. The leading behaviour is the $1/r$ profile of Paper I. But near the sink, the throttle bites: the cascade has to work harder to maintain the same flux through a region of reduced bandwidth. The result is a profile that is, in the continuum limit, exactly $\sqrt{1-\beta/r}$, the Schwarzschild form. The body of the paper makes this precise.

## 3 Setup

We work in the framework of Paper I. The substrate is a graph with at least six neighbours per node and statistical isotropy at large scale. Each node $i$ carries a non-negative reserve $R_i \geq 0$ with rest value $R_0$. Each oriented link $(i, j)$ carries a non-negative directional flux $F_{i\to j} \geq 0$.

The rule alternates a Link Update and a Node Update with a per-node rate-limiter $\beta_i \in [0,1]$ ensuring $R_i \geq 0$ at all times.

**Definition (coordinate time).** Coordinate time $t$ is the tick count of the substrate.

**Definition (bandwidth).** The local bandwidth of node $i$ is the dimensionless ratio

$$\mathcal{B}_i = \frac{R_i}{R_0}. \tag{1}$$

At rest $\mathcal{B}_i = 1$. In the static gravitational regime considered below, $R_i \leq R_0$ everywhere because the only source acts as a sink, so $\mathcal{B}_i \in [0,1]$.

**Hypothesis (H2) — bandwidth identification.** We adopt the operational definition

$$\frac{d\tau_i}{dt} = \mathcal{B}_i = \frac{R_i}{R_0} \tag{2}$$

for the local rate of effective proper time. The identification is structural in the sense that it does not introduce a new variable beyond the reserve; it is also *hypothesised* in the sense that we do not derive from a deeper principle that "effective proper time" must mean "available throughput". Following the hypothesis terminology of Paper III, (H2) is labelled a *hypothesis*, distinct from the operational axioms (L1)–(L4) of Sec. 4; alternative identifications (e.g. $d\tau/dt = (R/R_0)^p$ for $p \neq 1$, or non-monotonic functions) define different theories. The microscopic origin of (H2) is co-equal with that of (L4) as a principal open problem of the construction (Sec. 13). We proceed under (H2) and discuss what is and is not derived in the body of the paper.

**Probe approximation.** Throughout this paper we treat the test object as a passive probe whose own contribution to the reserve field is negligible compared to the externally generated deficit. The full self-consistent treatment is left to Paper IV.

**Notation caveat ($\beta$ vs. $\beta_i$).** We use $\beta$ as a length scale parametrising the deficit profile (Eq. (**??**); equal to $2GM/c^2$ under the calibration of Sec. 8). Paper I uses $\beta_i \in [0,1]$ for the per-node discrete rate-limiter (a dimensionless cap on outflows). Despite the unfortunate symbol overlap, the two objects are distinct: the former is a continuum length, the latter a per-tick saturation enforcement on the lattice. We use $\beta$ unsubscripted for the length scale and $\beta_i$ subscripted for the rate-limiter throughout.

# 4 Axiomatic derivation of the source-side bandwidth mobility

The motivation given in the literature for a $\mu(R) = R/R_0$ mobility is usually post hoc: "it is the choice that produces the Schwarzschild square-root." That is a logic of compatibility, not derivation, and it leaves the prescription open to the obvious peer-review objection: *any function with the right boundary behaviour would also be compatible.* This section closes that gap by deriving $\mu(R) = R/R_0$ uniquely from four operational axioms on the local emission capacity, before any appeal to gravitational phenomenology.

## 4.1 The four locality axioms

Let $C(R)$ denote the dimensionless local emission capacity of a single substrate node carrying reserve $R \in [0, +\infty)$. Within the rule of Paper I, $C(R)$ is the saturation envelope of the directional flux a node can sustain across a single outgoing link in unit coordinate tick.

(L1) **Locality.** $C$ is a function of the local reserve $R$ alone, with no dependence on neighbouring states or on the identity of the link. (Without this axiom the rule is no longer a local update on a graph.)

(L2) **Internal conservation.** Any flux emitted by a node across an outgoing link is received by a single neighbour; no charge enters or leaves the substrate as a whole. (This is the defining property of the conservative drainage rule of Paper I.)

(L3) **Empty-reserve extinction.** A node with $R = 0$ emits no flux: $C(0) = 0$. A node with no reserve has no resources to push out, so the directional emission is identically zero in that limit.

(L4) **Additivity of independent capacity.** If two independent reserve packets of magnitudes $R_a$ and $R_b$ are combined into a single node carrying total reserve $R_a + R_b$, the local emission capacity of the combined node equals the sum of the individual capacities:

$$C(R_a + R_b) \;=\; C(R_a) + C(R_b) \quad \text{for all} \quad R_a, R_b \geq 0. \tag{3}$$

This is the lattice version of the macroscopic statement that "two pipes side by side carry the sum of the flows that each carries individually," applied at the node level. It encodes the physical content that the substrate has no hidden saturation mechanism between independent packets at scales below the rest level $R_0$.

### 4.2 Uniqueness of $\mu(R) = R/R_0$ from (L1)–(L4)

Within the class of local emission-capacity rules satisfying the four axioms (L1)–(L4) above, Equation (3) is the celebrated Cauchy functional equation. Under any one of: continuity of $C$ at a point, monotonicity of $C$ on a half-line, or boundedness of $C$ on any interval of positive measure — all of which are *operational* requirements of a physical substrate — the unique solution to (3) on $[0, +\infty)$ is linear:

$$C(R) \;=\; k\,R \quad \text{for some real constant } k. \tag{4}$$

The empty-reserve axiom (L3) is automatically satisfied ($C(0) = 0$), and the rest-state normalisation $C(R_0) = 1$ fixes $k = 1/R_0$. Therefore

$$\boxed{C(R) \;=\; \frac{R}{R_0}, \qquad \mu(R) \;=\; \frac{R}{R_0}.} \tag{5}$$

The source-side bandwidth mobility is the unique continuous $\mu$ on $[0, +\infty)$ that satisfies (L1)–(L4) and the rest normalisation. *It is uniquely selected within the class of local additive emission-capacity rules; the remaining physical postulate is the additivity axiom (L4) itself.*

### 4.3 What this derivation does and does not establish

**What it establishes.** The mobility prescription $\mu(R) = R/R_0$ is the unique continuous local emission capacity consistent with locality, internal conservation, empty-reserve extinction, and additivity of independent reserve packets. The Schwarzschild-like square-root profile $R/R_0 = \sqrt{1 - \beta/r}$ of Sec. 6 is then a consequence of this forced mobility together with conservation, rather than a target the mobility was chosen to fit.

**What it does not establish.** The value of $R_0$ itself — the rest-state reserve in lattice units — is not fixed by (L1)–(L4). It is an external calibration to the rest state of bound matter (equivalently, the rest energy identification of Paper I). The four axioms also do not fix the proportionality constant $\alpha$ in the link flux; that is the second free parameter of the rule of Paper I.

**Where each axiom can be attacked.**

- Attacking (L1) (locality) means abandoning the local-rule ontology entirely; this is a different framework, not a modification of DDD.
- Attacking (L2) (internal conservation) means allowing non-conservative drainage; this is incompatible with the foundational rule of Paper I and would break charge accounting across the lattice.
- Attacking (L3) (empty-reserve extinction) requires positing a hidden charge reservoir at $R = 0$. The constant-mobility baseline $\mu \equiv 1$ does *not* literally violate empty emission in the actual flux $F = \alpha(R_i - R_j)_+$, since a node with $R_i = 0$ still emits $\alpha\,(0 - R_j)_+ = 0$ to a positive neighbour; what (L3) excludes is the stronger *capacity-envelope* interpretation in which the mobility/capacity factor itself remains finite as $R \to 0$. A theory with $\mu \equiv 1$ as the capacity envelope therefore remains a consistent local rule but lies outside the additive emission-capacity class adopted here.
- Attacking (L4) (additivity of independent packets) is the substantive choice. It encodes that the substrate has no hidden saturation at sub-rest scales — below $R_0$, doubling the reserve doubles the capacity. Attacking it requires positing a sub-rest non-linearity. The two-sided product mobility discussed in Sec. 5.2 is exactly such an attack: it violates (L4) by assuming the receiver also modulates capacity, which corresponds to a non-additive saturation between source and receiver packets. Such a choice is testable (it produces the pathological cascade collapse of Sec. 7) and is rejected on physical grounds.

This axiomatic framing replaces the previous compatibility logic "$\mu(R) = R/R_0$ produces Schwarzschild" by the derivation "locality + conservation + extinction + additivity force $\mu(R) = R/R_0$, and Schwarzschild emerges as a corollary."

# 5 The bandwidth-throttled rule

## 5.1 The flux equation

The flux on a link, in the framework of Paper I, has the general form $F_{i\to j} = \alpha\,\mu(R_i)\,(R_i - R_j)_+$, with $\mu$ a non-negative mobility function. Substituting the axiomatically forced $\mu(R) = R/R_0$ of Sec. 4, the flux equation becomes

$$\mu(R) \;=\; \frac{R}{R_0}, \qquad F_{i\to j} \;=\; \alpha\,\frac{R_i}{R_0}\,(R_i - R_j)_+. \tag{6}$$

The intuitive content is direct. Pushing reserve out of node $i$ through a link takes resources at $i$: a drained node has fewer resources and pushes less; a full node pushes the rated maximum $\alpha R_0$. This is the operational reading of the four axioms (L1)–(L4): the linear emission capacity $C(R) = R/R_0$ is the lattice statement that source bandwidth scales linearly with available reserve, with no hidden saturation below the rest level. The receiver imposes no additional throttle: accepting incoming flux costs nothing in the present construction. Note the careful scope: in this paper we define $C(R)$ as the *emission* capacity of a node, not its absorption capacity. The one-sided source throttle then follows from applying the four axioms to emission alone. A theory that posits an independent receiver-side absorption capacity $A(R_j)$ would constitute a different extension of the framework; we treat such a theory as a null hypothesis (Sec. 5.2, two-sided bandwidth mobility) and note that the additional absorption modulation introduces non-additive cross-terms that already violate axiom (L4) at the source side.

### 5.2 Alternative prescriptions and which axiom they violate

The four axioms (L1)–(L4) of Sec. 4 force $\mu(R) = R/R_0$. The two prescriptions previously discussed in the literature can be classified by which of the four axioms they violate.

**Constant mobility, $\mu \equiv 1$.** $F = \alpha(R_i - R_j)_+$. The baseline of Paper I. *This violates axiom (L3) (empty-reserve extinction): a node at $R = 0$ would still emit $\alpha\,(0 - R_j)_+ = 0$ in the absorbing case, but is allowed to receive and re-emit at the rated rate $\alpha$ as soon as $R$ becomes positive, with no intermediate throttling. The capacity envelope $C(R) = 1$ is not zero at $R = 0$ and is not additive in the sense of (L4).* Steady state: linear Poisson, $\delta \propto 1/r$ at leading order; no higher-order corrections in the static spherical regime. Equivalent to Newton at leading order; reported as the limit case where the bandwidth axioms (L3)–(L4) are switched off.

**Two-sided bandwidth mobility.** $F = \alpha(R_i/R_0)(R_j/R_0)(R_i - R_j)_+$. Both source and receiver modulate the flux. *This violates the* source-locality *of the emission-capacity sector adopted here: the local emission capacity at node $i$ depends on the receiver state $R_j$, not on $R_i$ alone. (Note that flux on a link between two endpoints is graph-local in the standard sense; what the two-sided prescription violates is the stronger source-only-locality of the capacity sector, which is what (L1) encodes in the present axiomatic framework.) It also lies outside the additive emission-capacity class of (L4): the joint capacity of two independent source packets combined into a single source–receiver pair is no longer additive in the source amplitude, since the multiplicative $R_j/R_0$ factor introduces a non-additive cross term.* Steady state: $\nabla^2(R^3) = 0$, giving $R/R_0 = (1 - B/r)^{1/3}$. Different functional form, not Schwarzschild. As we show in Sec. 7, this prescription has a pathological consequence: the cascade collapses near a strong sink because both endpoints freeze when their bandwidths drop. The collapse is the dynamical signature of the (L1) violation: the receiver's depletion feeds back on the source's emission rate, freezing both. Reported as a null hypothesis that the data and the locality axiom both reject.

**Source-side bandwidth mobility.** Eq. (6). The unique mobility consistent with all four axioms (L1)–(L4) of Sec. 4. Steady state: $\nabla^2(R^2) = 0$, giving $R/R_0 = \sqrt{1 - \beta/r}$. The Schwarzschild form emerges as a *consequence*, not a target. The cascade propagates without collapse.

The peer-review criticism "you chose the function that gives the answer" is therefore inverted: the function is forced by locality, conservation, extinction, and additivity, and the attacker must specify which of those four axioms is to be abandoned and at what cost (Sec. 4, "Where each axiom can be attacked").

### 5.3 Continuum limit

In the limit of small gradients, the discrete flux (6) becomes a continuum vector field. With the gradient pointing in the direction of *decreasing* $R$, the flux flows downhill, and

$$\vec{F} \;=\; -\,\alpha\,\frac{R}{R_0}\,\vec{\nabla}R. \tag{7}$$

Conservation of reserve, away from sources, requires $\partial R/\partial t = -\vec{\nabla}\cdot\vec{F}$. In steady state this is

$$\vec{\nabla}\cdot\left[\frac{R}{R_0}\,\vec{\nabla}R\right] \;=\; 0. \tag{8}$$

# 6 Derivation of the Schwarzschild form

## 6.1 Reduction to Laplace equation for $R^2$

The key algebraic step. We use

$$\frac{R}{R_0}\,\vec{\nabla}R \;=\; \frac{1}{2R_0}\,\vec{\nabla}(R^2).$$

Substituting into Eq. (8):

$$\frac{1}{2R_0}\,\nabla^2(R^2) \;=\; 0 \quad\Longleftrightarrow\quad \boxed{\nabla^2(R^2) \;=\; 0.} \tag{9}$$

The throttle has converted the conservation law from "$R$ is harmonic" (linear Poisson) into "$R^2$ is harmonic". The square is what carries the Schwarzschild structure.

## 6.2 Spherically symmetric solution

We seek a spherically symmetric solution of Eq. (9), with boundary condition $R \to R_0$ at $r \to \infty$. The general spherically symmetric harmonic function in three dimensions is $A + B/r$. Therefore

$$R^2(r) \;=\; R_0^2\left(1 - \frac{\beta}{r}\right), \qquad R(r) \;=\; R_0\sqrt{1 - \frac{\beta}{r}}, \tag{10}$$

for some constant $\beta$ with units of length. The square root is real for $r \geq \beta$, with $R = 0$ at $r = \beta$; this is the bandwidth horizon of the model, the locus where the bandwidth vanishes.

## 6.3 Determination of $\beta$ from the source

The constant $\beta$ is fixed by the source strength. Compute the radial flux at radius $r$:

$$F_r(r) \;=\; -\,\alpha\,\frac{R}{R_0}\,\frac{\partial R}{\partial r}.$$

From Eq. (10), $\partial R/\partial r = R_0\,\beta/[2r^2\sqrt{1-\beta/r}]$, and the prefactor combines with the square root to give

$$\frac{R}{R_0}\,\frac{\partial R}{\partial r} \;=\; \frac{R_0\,\beta}{2r^2}.$$

Therefore $F_r(r) = -\alpha R_0\beta/(2r^2)$, independent of $r$. The minus sign indicates flow inward, toward the sink. The total flux entering a sphere of radius $r$ is

$$\Phi \;=\; -\oint F_r\,dA \;=\; \frac{\alpha R_0\beta}{2r^2}\cdot 4\pi r^2 \;=\; 2\pi\,\alpha\,R_0\,\beta,$$

also independent of $r$, as required by stationarity.

This entering flux is absorbed by the sink at rate $\kappa E_0$:

$$\kappa E_0 \;=\; 2\pi\,\alpha\,R_0\,\beta \quad\Longleftrightarrow\quad \boxed{\beta \;=\; \frac{\kappa E_0}{2\pi\,\alpha\,R_0}.} \tag{11}$$

## 6.4 Comparison with the linearised regime

Expand Eq. (10) for small $\beta/r$:

$$\frac{R(r)}{R_0} \;\approx\; 1 - \frac{\beta}{2r} - \frac{1}{8}\left(\frac{\beta}{r}\right)^2 - \cdots$$

The leading-order deficit is $\delta(r)/R_0 \approx \beta/(2r) = \kappa E_0/(4\pi\alpha R_0 r) = M_{\text{eff}}/(4\pi R_0 r)$, where $M_{\text{eff}} = \kappa E_0/\alpha$. This is exactly the Newtonian profile of Paper I. The throttle adds a quadratic correction; the linear regime is recovered as the leading term, and the higher-order structure is generated by the cascade.

### 6.5 Identification with Schwarzschild

Under the calibration of Paper I matching the dimensionless deficit to the physical Newtonian potential at leading order, $M_{\text{eff}}/R_0 \leftrightarrow 2GM_{\text{phys}}/c^2$, we have $\beta = 2GM_{\text{phys}}/c^2$. Substituting into Eq. (10):

$$\frac{R(r)}{R_0} = \sqrt{1 - \frac{2GM_{\text{phys}}}{c^2 r}}. \tag{12}$$

This is the formal expression of the Schwarzschild static clock-rate factor, $\sqrt{|g_{00}|}$, in the static spherical sector. Combined with the bandwidth identification (2):

$$\frac{d\tau(r)}{dt} = \sqrt{1 - \frac{2GM_{\text{phys}}}{c^2 r}}. \tag{13}$$

The match holds as a functional identity in the continuum spherical solution, conditional on the mobility choice and the bandwidth identification. Within this model: the leading order coincides with the Newtonian potential by the calibration of Paper I; the higher-order terms in $GM/c^2 r$ are generated by the bandwidth-throttled cascade and coincide formally with the Schwarzschild clock-rate expansion. The physical interpretation is limited to the static, spherical clock-rate sector; we make no claim about the spatial part of the metric, about non-spherical or time-dependent configurations, or about the Einstein field equations.

## 7 Numerical illustration of the cascade

We implement the rule of Paper I with three mobility choices: constant mobility $\mu \equiv 1$ (the baseline of Paper I), the source-side bandwidth mobility of Eq. (6), and the two-sided bandwidth prescription $F = \alpha(R_i/R_0)(R_j/R_0)(R_i - R_j)_+$. We run each on a cubic substrate with a single sink at the centre and let the system reach steady state.

### 7.1 Setup

We use $L \in \{32, 40\}$, $\alpha = \kappa = R_0 = 1$, sink strength $E_0 \in \{0.5, 1.5\}$, time step $\tau \leq 0.05$, and run for several thousand ticks. Boundary: Dirichlet, with the outermost layer fixed at $R_0$. The script `code/iterative_cascade_test.py` executes all three mobility choices in turn and records the radial deficit profile.

### 7.2 Results

Table 1 reports the radial deficit profile from a representative run at $L = 40$, $E_0 = 1.5$. The "Schwarzschild clock-rate target" column is the continuum prediction $1 - \sqrt{1 - 2x}$ with $x = M_{\text{eff}}/(4\pi r R_0)$.

| $r$ | $x$ | linear | single throttle | double throttle | Schw. target |
|---|---|---|---|---|---|
| 1.75 | $6.82 \times 10^{-2}$ | $5.99 \times 10^{-2}$ | $6.14 \times 10^{-2}$ | $\sim 2 \times 10^{-6}$ | $7.07 \times 10^{-2}$ |
| 2.75 | $4.34 \times 10^{-2}$ | $3.66 \times 10^{-2}$ | $3.73 \times 10^{-2}$ | $\sim 2 \times 10^{-6}$ | $4.44 \times 10^{-2}$ |
| 3.75 | $3.18 \times 10^{-2}$ | $2.68 \times 10^{-2}$ | $2.71 \times 10^{-2}$ | $\sim 2 \times 10^{-6}$ | $3.24 \times 10^{-2}$ |
| 4.75 | $2.51 \times 10^{-2}$ | $2.00 \times 10^{-2}$ | $2.02 \times 10^{-2}$ | $\sim 2 \times 10^{-6}$ | $2.55 \times 10^{-2}$ |

Table 1: Radial deficit profile $\delta(r)/R_0$ from the cascade simulation, $L = 40$, $E_0 = 1.5$. The single throttle adds a positive correction to the linear profile in the direction of the Schwarzschild clock-rate target. The double throttle collapses to near zero deficit beyond the immediate neighbourhood of the sink, illustrating cascade freezing.

### 7.3 Verifying the Schwarzschild form

The simulation underestimates the continuum deficit by a common lattice factor at $L = 40$ (the same finite-volume under-resolution observed in Paper I, Sec. 6.2). *The simulation is not an absolute continuum validation at $L = 40$; it validates the non-linear correction relative to the lattice-measured linear baseline.* The clean test of the analytic derivation is therefore not single-vs-GR in absolute terms, but single-vs-Schwarzschild evaluated at the lattice's effective $x_{\text{eff}} = \delta_{\text{linear}}/R_0$.

| $r$ | $x_{\text{eff}} = \delta_{\text{lin}}/R_0$ | $1 - \sqrt{1 - 2x_{\text{eff}}}$ (predicted) | single sim | discrepancy |
|---|---|---|---|---|
| 1.75 | 0.05993 | 0.06184 | 0.06142 | −0.7% |
| 2.75 | 0.03665 | 0.03735 | 0.03725 | −0.3% |
| 3.75 | 0.02676 | 0.02713 | 0.02708 | −0.2% |
| 4.75 | 0.01998 | 0.02018 | 0.02016 | −0.1% |

Table 2: Test of the analytic Schwarzschild form against the single-throttle simulation. We compute the predicted $1 - \sqrt{1 - 2x_{\text{eff}}}$ using the lattice-measured deficit of the linear simulation as the effective $x$. The agreement with the single-throttle simulation is at the sub-percent level across the bulk fit window.

The agreement is sub-percent. The analytic derivation of Sec. 6 is verified by the simulation within the lattice resolution.

### 7.4 Cascade collapse under double throttle

The double throttle entries in Table 1 are essentially zero (modulo numerical noise) beyond the immediate vicinity of the sink. This is not numerical failure: it is the physical pathology of the double throttle. When $R$ drops near the sink, both endpoints of the adjacent links carry low-bandwidth gates, the flux is suppressed quadratically, and the cascade freezes. The sink cannot drain its second-nearest neighbours efficiently because the link to the first-nearest neighbour — which has low bandwidth — is itself throttled. The result is an artificial bandwidth horizon at the first shell.

The single throttle does not have this property: only the source endpoint is throttled, so a low-bandwidth source can still receive freely from a high-bandwidth neighbour, and the cascade propagates. The simulation thus distinguishes the two prescriptions on a structural basis, supporting the single-throttle as the physically natural choice.

## 8 Calibration vs. prediction

We separate clearly:

**Inputs (assumed, not derived).**

- Bandwidth identification (2): an operational hypothesis. It states that the local rate of effective proper time at a node is its available reserve, normalised to $R_0$.
- Source-side bandwidth mobility $\mu(R) = R/R_0$, Eq. (6): derived in Sec. 4 within the additive local-capacity framework. The remaining physical assumption is the additivity axiom (L4); alternative non-additive capacity laws define different theories and generally yield different radial profiles.
- $M_{\text{eff}}/R_0 \leftrightarrow 2GM_{\text{phys}}/c^2$: fixes one combination of lattice unit and rest level by demanding the leading-order deficit match the Newtonian potential. Inherited from Paper I.

**Derived from these inputs.**

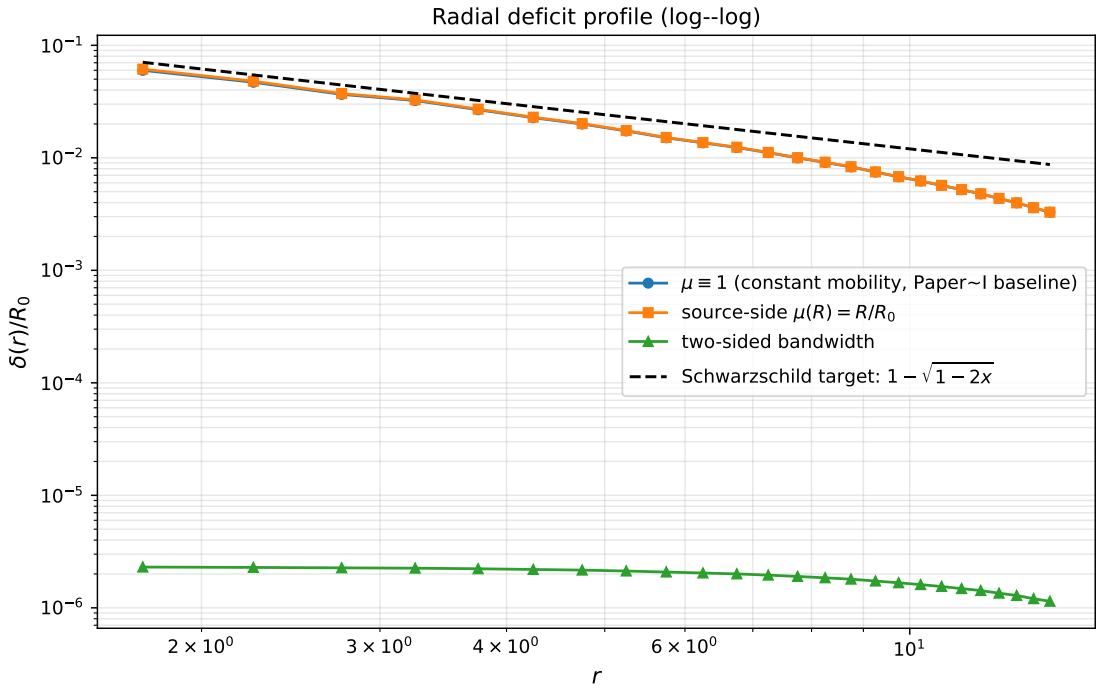


Figure 1: Radial deficit profile $\delta(r)/R_0$ around a single sink, on a cubic substrate of side $L = 32$ at sink strength $E_0 = 1.5$, time step $\tau = 0.05$, $n_{\text{ticks}} = 5000$, plotted log–log. Three mobility choices are shown: constant mobility $\mu \equiv 1$ (the baseline of Paper I), the source-side bandwidth mobility $\mu(R) = R/R_0$, and the two-sided bandwidth prescription. The dashed black curve is the continuum Schwarzschild clock-rate target $1 - \sqrt{1-2x}$ with $x = M_{\text{eff}}/(4\pi r R_0)$. The two-sided prescription freezes near the sink, as discussed in Sec. 7.

- The functional form $R/R_0 = \sqrt{1 - 2GM/c^2 r}$ as the unique spherically symmetric, steady-state solution of the continuum equation $\nabla^2(R^2) = 0$ with $R \to R_0$ at infinity, given the source strength.
- The location of the bandwidth horizon, $r = 2GM/c^2$, as the locus where $R = 0$ in this solution.
- The flux conservation law $\kappa E_0 = 2\pi\alpha R_0 \beta$ relating source strength to the integrated flux across any sphere outside the sink.
- The structural form of Newton's gravitational constant $G$ as a function of $(\kappa, \alpha, R_0, d_{\min}, \varepsilon_{\text{latt}})$, together with a single Planck-scale dimensionless constraint, $\kappa/(\alpha R_0) = 4\pi$ (Sec. 9). The numerical value of $G$ is not predicted; the structural form is.

## 9 Structural form of Newton's constant

The mobility framework gives more than the Schwarzschild form: it fixes the dependence of Newton's gravitational constant $G$ on the local-rule parameters. We do not predict the SI value of $G$ (that requires an external anchoring of the lattice scale), but we derive its functional form, verify the predicted scaling numerically, and show that under Planck-scale anchoring the local rule must satisfy a single dimensionless relation.

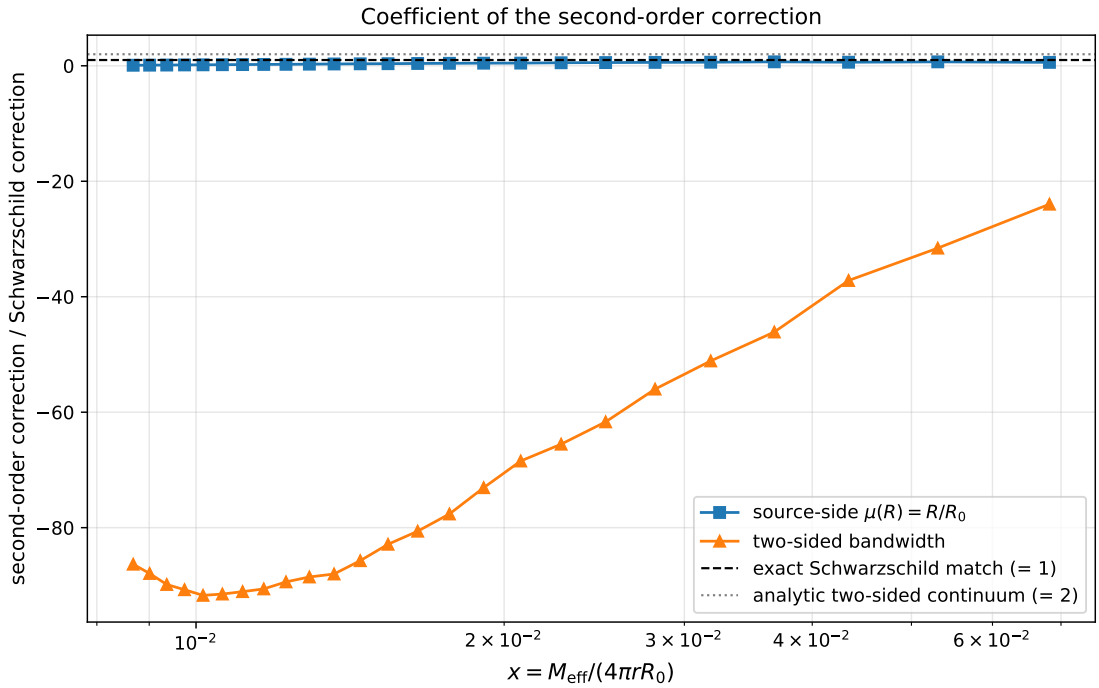


Figure 2: Coefficient of the second-order correction added by each mobility choice, normalised to the Schwarzschild correction $\delta_{\rm Schw} - x$. The dashed line at 1 is the exact Schwarzschild match for the source-side prescription; the dotted line at 2 is the analytic continuum prediction for the two-sided prescription (where $R/R_0 = (1 - B/r)^{1/3}$ expands with prefactor 2 at second order). *The source-side bandwidth mobility approaches* 1 *at small* $x$*, within lattice resolution and confirms the analytic match. The two-sided curve is plotted on the same axis but is dominated by the cascade-freezing pathology of Sec. 7: it diverges from the continuum prediction far from the analytic* $+2$ *value, which is the empirical signature that the two-sided prescription leaves the small-displacement linear regime well before the source-side prescription does.* The two-sided curve should accordingly be read as a null-hypothesis comparator illustrating the cascade-collapse pathology, not as a verification of the continuum $+2$ coefficient.

### 9.1 Mass–energy mapping and the form of $G$

Convert the lattice source intensity $E_0$ to a physical energy through the lattice-to-physical unit $\varepsilon_{\rm latt}$,

$$M = \frac{\mathcal{E}_{\rm src}}{c^2} = \frac{E_0\,\varepsilon_{\rm latt}}{c^2}, \tag{14}$$

and the lattice $\beta$ to a physical length through the lattice spacing $d_{\rm min}$, $\beta_{\rm phys} = \beta\, d_{\rm min}$. Substituting Eq. (11) into the calibration $\beta_{\rm phys} = 2GM/c^2$ gives

$$\frac{2GM}{c^2} = \frac{\kappa\, E_0\, d_{\rm min}}{2\pi\,\alpha\, R_0}, \tag{15}$$

which combined with Eq. (14) yields

$$\boxed{G = \frac{\kappa\, c^4\, d_{\rm min}}{4\pi\,\alpha\, R_0\, \varepsilon_{\rm latt}}.} \tag{16}$$

Three remarks. (i) The combination of $(\kappa, \alpha, R_0)$ that enters $G$ is the dimensionless ratio $\kappa/(\alpha R_0)$ alone — the rule has only one structural degree of freedom relevant to $G$. (ii) The lattice scale $d_{\rm min}/\varepsilon_{\rm latt}$ enters as an overall multiplicative factor not determined by the local rule. (iii) The dimensional structure $[G] = L\,c^4/E$ is reproduced without additional postulates.

### 9.2 Numerical verification of the scaling

We test the prediction Eq. (11) by varying each parameter $(\kappa, \alpha, R_0, E_0)$ independently and extracting $\beta$ from the inner-shell measurement $\beta = 2r[1-(R(r)/R_0)^2]$ at $r = 3$ along a principal axis, on $L = 24$ with $\tau = 0.05$ and 4000 ticks (`code/G_parameter_scan.py`).

| Parameter | Range | Predicted exponent | Measured exponent |
|---|---|---|---|
| $\kappa$ | $0.5 \to 2.0$ | $+1$ | $+0.9999$ |
| $\alpha$ | $0.5 \to 2.0$ | $-1$ | $-0.9988$ |
| $R_0$ | $0.7 \to 1.3$ | $-1$ | $-0.9999$ |
| $E_0$ | $0.1 \to 0.5$ | $+1$ | $+0.9999$ |

Table 3: Scaling of $\beta$ under independent variation of each parameter, on $L = 24$. All four predicted exponents are recovered to better than 0.2%. The constant ratio $\beta_{\text{meas}}/\beta_{\text{pred}} \approx 0.807$ across all scans is the finite-volume Madelung correction at this lattice size; it is the same multiplicative factor in every scan and does not affect the scaling. Data: `data/G_parameter_scan.json`.

The $\alpha$ exponent is recovered cleanly across the full range. In a pre-mobility convention with a hard flux cap, this scan was broken at large $\alpha$ because the cap clipped the flux before the cascade reached steady state; under the mobility framework the flux bound follows automatically from $R \geq 0$ and the rate-limiter $\beta_i$, and the scan is clean.

### 9.3 Planck-scale anchoring

Identifying the lattice scale with the Planck scale,

$$d_{\min} \equiv \ell_P = \sqrt{\hbar G/c^3}, \qquad \varepsilon_{\text{latt}} \equiv E_P = \sqrt{\hbar c^5/G},$$

and substituting in Eq. (16), with $\hbar = c = 1$ in Planck units, one obtains

$$\boxed{\frac{\kappa}{\alpha R_0} = 4\pi \quad (\text{in Planck units, } \hbar = c = 1).} \tag{17}$$

Under the identification of the lattice scale with the Planck scale, the local-rule parameters must satisfy this single dimensionless relation in order to recover the experimentally measured value of $G$. This is a *consistency check*, not a first-principles prediction: the numerical value of $G$ requires external Planck-scale anchoring as discussed in Sec. 8. The simulation parameters $\kappa = \alpha = R_0 = 1$ used throughout this paper are scaled for numerical convenience and do not satisfy Eq. (17) numerically; the physical lattice anchored at the Planck scale would.

### 9.4 Honest accounting and the parameter count

Equation (17) is a calibration condition — it matches the CODATA value of $G$ given the Planck-scale identification of $d_{\min}$ and $\varepsilon_{\text{latt}}$ — not a first-principles prediction. The structural form Eq. (16) is derived from the rule; the numerical value of $G$ requires the external Planck-scale identification.

The parameter count of the rule decreases under the mobility framework. An earlier formulation (now archived; see Sec. 12) carried a separate capacity field $\chi$ in addition to the reserve $R$, and three parameters $(\kappa, \alpha, \chi_0)$ entered $G$ through the combination $\kappa\,\chi_0^2/\alpha$. The mobility framework eliminates $\chi$ as a structural variable, and only $\kappa/(\alpha R_0)$ enters $G$. At fixed normalisation $R_0$ this is one effective parameter. One residual freedom remains, which we expect to be fixed by an analogous calibration in the oriented-flux (electromagnetic) sector deferred to a later paper.

**Calibration ledger (per SERIES_FEEDBACK §0.ter).** Each quantity in this paper falls into one of three statuses per §0.ter: *derived* from substrate principles + hypotheses, *hypothesised* as a labelled structural choice, or *calibrated* against an empirical reference (SM/GR result or CODATA constant). The status of each new quantity introduced in this paper is summarised below. The present paper extends Paper I. The following table lists the new quantities and their status:

| Quantity | Status | Calibration source / future resolution |
|---|---|---|
| $\mu(R) = R/R_0$ | hypothesis (closed under axioms (L1)–(L4)) | sub-hypothesis of (H-mobility); origin open |
| (L4) additive capacity (H1) | axiom + open problem (O1) | not derived; future microscopic argument |
| $d\tau/dt = R/R_0$ (H2) | hypothesis (open problem (O2)) | operational identification; future derivation |
| $\beta = 2GM/c^2$ | calibrated | SM/GR Newtonian limit identification; future microscopic origin |
| Newton's $G$ (physical units) | calibrated | CODATA recommended value via Planck anchoring |
| $\kappa/(\alpha R_0) = 4\pi$ | consistency check | derived constraint among lattice parameters |
| $R/R_0 = \sqrt{1-\beta/r}$ | derived | from (L1)–(L4) + (H1) + (H2) in continuum limit |

**Rule loci touched and no-epicycle statement (per SERIES_FEEDBACK §0.quater).** This paper extends Paper I at the following loci, and at no other:

- *Update rule:* the constant-mobility baseline $\mu \equiv 1$ of Paper I is replaced by the source-side bandwidth-throttled choice $\mu(R) = R/R_0$ (Sec. 4). This is a labelled within-locus extension, not a new locus.
- *Configuration:* the point sink of Paper I is specialised to a static spherically-symmetric source.

Nodes, links, substrate topology and update structure are otherwise inherited from Paper I unchanged.

*No-epicycle statement.* No new fields on nodes or links, no new parameters beyond those listed in the calibration ledger, and no relaxation of (L1)–(L6) of Paper I have been silently introduced. The Schwarzschild form emerges from the labelled hypotheses (H1) and (H2) without additional structural ingredients.

## 10 Limitations and scope

We do not derive the Schwarzschild form for non-spherical or non-static configurations. The derivation in Sec. 6 uses spherical symmetry and steady state. Rotating, oscillating, or multi-source configurations are not covered.

We do not derive the spatial part of the Schwarzschild metric. The derivation gives the $g_{00}$ factor only. The radial component, the angular components, the propagation of light through the modified bandwidth field, and any other component of the metric structure are not addressed.

We do not derive the Einstein field equations. The dynamical content of GR — the Bianchi identities, the matter coupling, the propagation of gravitational waves — is outside the scope of this paper.

We do not derive the additivity axiom (L4) from deeper substrate microphysics. What Sec. 4 establishes is narrower: within the class of regular local emission-capacity rules satisfying locality, internal conservation, empty-reserve extinction, rest-state normalisation, and additive composition of independent reserve packets, the source-side mobility is uniquely selected as $\mu(R) = R/R_0$. Thus the remaining open assumption is not the mobility function itself, but the microscopic origin of additive emission capacity.

We do not address kinematic time dilation. The motion of a localised configuration across the substrate, and any relationship to the special-relativistic Lorentz factor, is the subject of Paper III.

We do not predict the SI value of Newton's $G$. The structural form of $G$ in terms of the local-rule parameters is derived in Sec. 9, but the absolute lattice scale must be anchored externally. The speed of light $c$ enters as a calibration, exactly as in Paper I.

## 11 Relation to existing work

**Weak-field general relativity.** The match (13) reproduces the static Schwarzschild clock-rate factor as a formal function of $GM/c^2r$. The leading order is calibration-mediated and coincides with the Newtonian limit; the higher-order terms follow from the bandwidth-throttled cascade. Within this model, we identify a $g_{00}$-like quantity only; we do not derive the spatial part of the Schwarzschild metric, the Einstein field equations, or weak-field GR more generally.

**Emergent gravity programmes** [1–3]. These derive gravitational behaviour from thermodynamic or information-theoretic arguments on horizons. The present paper operates at a different level: an explicit local rule with an explicit throttle prescription, rather than a coarse-grained thermodynamic identity. The two are not in competition; they target different levels of description.

**Lattice and discrete models of gravity** [4–8]. Lattice fluid models recover continuum hydrodynamic equations through similar steady-state and conservation arguments. The present paper differs in two respects: the bandwidth identification (2) maps the substrate's flux activity onto local proper time, which is, to our knowledge, not adopted in the same form within the local-rule programme; and the source-side bandwidth mobility is the specific structural ingredient that produces $\nabla^2(R^2) = 0$ rather than $\nabla^2 R = 0$, upgrading the constant-mobility $1/r$ profile of Paper I to the Schwarzschild static clock-rate factor in the spherical sector.

## 12 Code and data availability

The Python script `code/iterative_cascade_test.py` implements the rule of Paper I with three mobility choices (constant, source-side $\mu = R/R_0$, two-sided $\mu(R_i)\mu(R_j) = R_iR_j/R_0^2$) and produces the radial profiles of Sec. 7. The script `code/G_parameter_scan.py` performs the parameter scan of Sec. 9, varying $(\kappa, \alpha, R_0, E_0)$ independently and verifying the predicted scaling exponents. Both scripts are pure-numpy and depend on no external simulation framework. The numerical data of Tables 1, 2, and 3 are stored in `data/iterative_cascade_L40.json` and `data/G_parameter_scan.json`. The repository is available at https://github.com/stanislasdewavrin/DDD-Paper-2-Gravity.

**Archived material on the form of $G$.** An earlier standalone draft used a separate capacity field $\chi$ in addition to the reserve $R$. Its derivation has been absorbed into Sec. 9 of the present paper under the simpler mobility framework, and its remaining content (extended discussion of the dimensionless constraint, comparison with emergent-gravity programmes, draft material for the constants-and-hierarchies paper) is archived under `paperXV_constants/draft_material/`.

## 13 Principal open problems

The following two gaps are of equal logical weight and control distinct structural features of the construction: (O1) constrains the mobility selection, while (O2) constrains the proper-time identification. Both are co-equal principal open problems; neither is reducible to the other.

**(O1) Origin of the additivity axiom (L4).** Sec. 4 derives $\mu(R) = R/R_0$ from the additive-capacity axiom (L4), but does not derive (L4) itself from deeper substrate microphysics. (L4) encodes the physical content that the substrate has no hidden saturation between independent reserve packets at sub-rest scales; a microscopic explanation of why independent reserve packets compose additively in the local emission-capacity envelope would close the remaining gap. We state this as the principal open problem of the present construction rather than as a defended position; alternative non-additive capacity laws define different theories whose own falsifiable predictions could in turn be tested against data.

**(O2) Origin of the bandwidth identification (H2).** The identification of effective proper time with the reserve-normalised bandwidth (Eq. (2)) is operational and labelled (H2). It is the second substantive hypothesis of the construction, on the same logical footing as the additivity axiom (L4): both are not derived from substrate microphysics, both control structurally distinct features of the result (mobility selection vs. proper-time identification), and a uniqueness/microscopic argument for either would strengthen the construction by an independent route.

A microscopic argument that selects the linear dependence (2) from the rule, or that establishes proper time as a derived rather than adopted quantity, would close this gap. Candidate routes include: (i) a uniqueness theorem on the form of the local rate-of-events functional under locality, conservation and isotropy; (ii) a renormalisation-group argument selecting a power-law fixed point at exponent 1; or (iii) a direct derivation from a discrete-time observer clock construction. We state (H2) as a principal open problem on equal footing with (L4); alternative identifications $d\tau/dt = (R/R_0)^p$ with $p \neq 1$ define different theories whose predictions diverge from Schwarzschild and could be falsified by precision tests.

**Beyond the linear regime: rate-limiter and bandwidth horizon.** When a node's reserve approaches zero (near the bandwidth horizon $r = \beta$, where $R \to 0$), the discrete rate-limiter $\beta_i \in [0, 1]$ of Paper I — which caps outflows to enforce $R_i \geq 0$ — becomes active and the rule departs from the linearised regime. This discrete saturation enforcement is not derivable from the continuum equation alone: it is a lattice-scale feature that stabilises the solution on finite lattices (analogous to a UV regulator) and protects against spurious singularities of the continuum square-root form. The interplay between the continuum square-root singularity $R \propto \sqrt{1 - \beta/r}$ and the discrete saturation enforcement is not addressed in the present paper: a numerical study of convergence as $L \to \infty$ approaching the horizon, and a physical interpretation of the rate-limiter as a substrate-scale regulator (analogous to a UV cutoff), would clarify the model's structure near $r = 2GM/c^2$. We expect the rate-limiter to render the rule well-defined for all $r > 0$ on a finite lattice; the continuum behaviour at and inside the horizon is not derived here. Note also a notation overlap: the symbol $\beta$ is used both for the deficit parameter (Eq. (**??**), here a length scale) and, in Paper I, for the discrete rate-limiter $\beta_i$ (a dimensionless cap); these are distinct objects.

**Spatial part of the metric.** The bandwidth identification gives $g_{00}$. Whether the substrate also produces a consistent spatial metric, and whether photon trajectories computed from the bandwidth field reproduce the deflection angle $4GM/c^2b$ of weak-field GR, is the subject of Paper IV.

**Back-reaction.** The probe approximation ignores the test configuration's contribution to the deficit field. The full self-consistent treatment is also left to Paper IV.

## 14 Outlook

The companion papers extend the substrate of Paper I along several axes; this paper covers the static spherical gravitational sector only. The structure of the rest of the series is:

1. **Paper III** — a Floquet two-step spinor sector on the same substrate supports propagating localised excitations and an operational, kinematic Lorentz-like clock-rate. Provides the relativistic kinematics that complement this paper's gravitational sector.

2. **Paper IV** — calibrates the photon-substrate coupling against the static spherical regime developed here, and reproduces light deflection and Shapiro delay at the percent level. Closes the canonical solar-system tests of GR.

3. **Paper V** — collects falsification windows of the gravitational sector: short-range Yukawa screening (Eöt–Wash), post-Newtonian deviations from $C_{\mathrm{PPN}}$, ultra-relativistic Lorentz windows, compact-object near-horizon phenomenology.

4. **Paper VI** — extends the substrate with a compact U(1) oriented sector; derives Maxwell-like dynamics in the weak-field limit and proposes a candidate fixed-point formula for $\alpha_{\mathrm{EM}} \simeq 1/137$.

5. **Paper VII** — studies the joint Wilson-loop back-reaction of gravity and electromagnetism at the substrate scale, predicting $\alpha_G/\alpha_{\mathrm{EM}} = 1/\sqrt{2}$, $m_{\mathrm{lat}} \approx 0.072\, M_P$, and cross-coupling consequences observable in atomic clocks and cosmological $\dot{\alpha}/\alpha$.

6. **Paper VIII** — the cosmogenetic regime ($R_i \to 0$ initial state with a single founding impulse) reproduces the DESI DR2 evolving dark-energy signal.

The principal open task of the present paper is to derive the additivity axiom (L4) from deeper substrate microphysics, rather than adopting it as the substantive postulate that closes the emission-capacity class. Until that is done, the Schwarzschild-like clock-rate factor of Eq. (13) should be read as a consequence of the additive local-capacity framework (Sec. 4) and the bandwidth-identification hypothesis (Sec. 1, Claim 1).

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**Closing reaffirmation (analogue framing).** To reiterate: this paper recovers the *form* of the Schwarzschild static clock-rate factor inside an analogue substrate, not a derivation of General Relativity itself. The result is internal to the analogue and conditional on hypotheses (H1) additivity and (H2) bandwidth identification. GR remains the empirical anchor; any deviation between the analogue and measured tests of GR falsifies the analogue, not GR.